

Battery → DC–DC Converter → DC Bus (Hybrid Ferry Front-End)

In this task, you will build a straightforward power electronics chain used in hybrid electric ferries.

You will model only the front section of the ferry's electrical system:

Battery → DC–DC Converter → DC Bus

Your goal is to create a working Simulink/Simscape model that takes power from a battery, regulates it using a DC–DC converter, and supplies a stable DC bus voltage.

Your Model Must Contain:

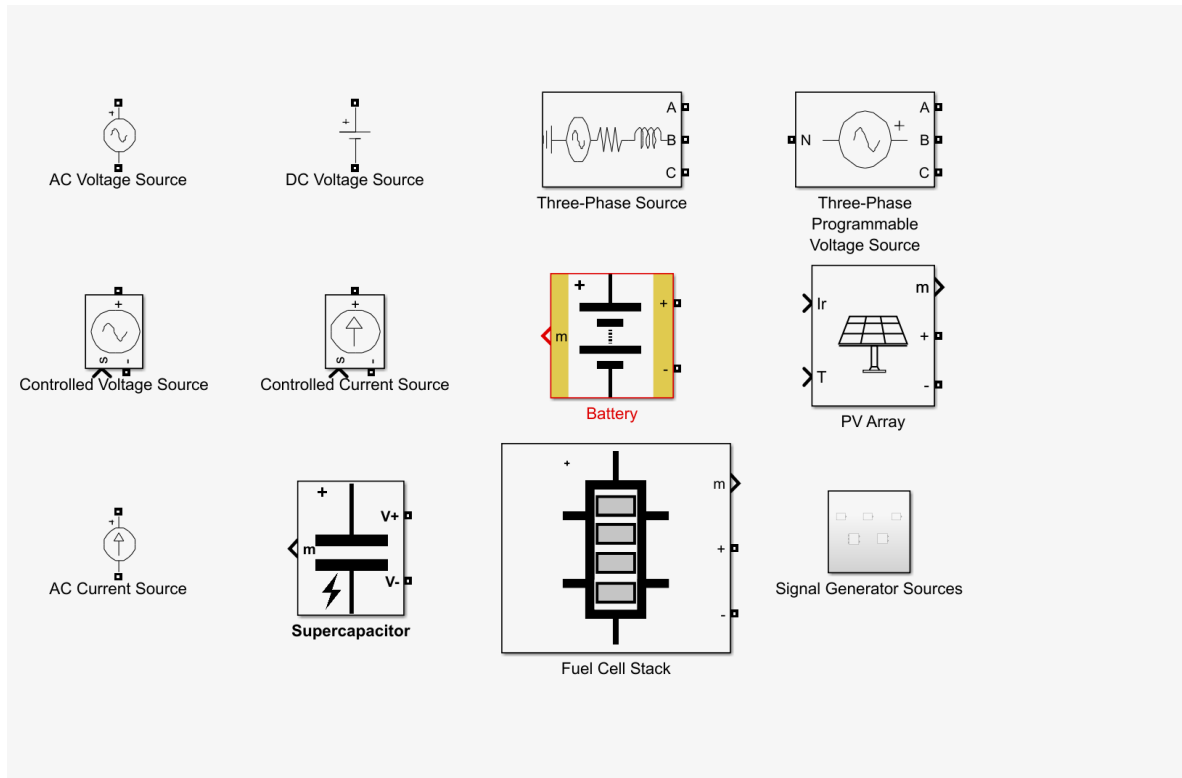
Battery (DC Source)

Use the built-in Simscape block:

- **DC Voltage Source**
(*Simscape → Electrical → Specialized Power Systems → Sources*)

sps_lib/Sources:

Simulation Task (Upload Only Your .slx File)



Set any reasonable voltage (e.g., 100–250 V).
Label it **Battery**.

DC–DC Converter:

You can use:

You may choose **either** a Buck converter **or** a Boost converter.

MATLAB already provides complete, working models.

<https://www.mathworks.com/help/sps/ug/buck-converter-voltage-control.html>

<https://www.mathworks.com/help/sps/ug/boost-converter-voltage-control.html>

What you do with the example:

- Copy the converter subsystem (the whole circuit + controller + PWM).
- Paste it into your own new model.
- Connect **Battery** → **Converter** → **DC Bus**.
- Adjust:
 - Battery voltage
 - Converter output voltage target
 - (Optional) L and C values (small changes only)

Simulation Task (Upload Only Your .slx File)

You do **not** need to modify the PI controller or PWM unless you want to.

DC Bus (Very Simple):

At the converter output, place:

- **One Capacitor**
- **One Resistor** (load)

This represents the ferry's DC bus.

Your converter must regulate the DC bus voltage.

You should be able to see:

- Output voltage settling to the reference
- Inductor current (from the example)
- Battery → Converter → DC Bus connection

Your Simulink model: `<group_Number>_ferry.slx`